

ANAV VAIDYA

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Systems Engineering senior seeking 2026/2027 internship or full-time opportunities.

Education

University of Illinois Urbana-Champaign

B.S. Systems Engineering & Design | Focus: Robotics & Control Systems

Minor: Electrical Engineering

Expected May 2027

Urbana-Champaign, IL

Relevant Coursework

Controls & Robotics: Robotics · Control Systems (Feedback, State-Space, Digital)

Mechanical Modeling & Design: Design of Structures and Mechanisms · Solid Mechanics · Dynamics

Other: Power Circuits · Optimization Models · Analog Signal Processing

Technical Skills

Robotics & Controls: ROS1 · UR3 manipulator · OpenCV · Matlab/Simulink · NI LabVIEW

Programming: Python · C/C++ · SQL · Linux

Embedded Systems & Electronics: Arduino · TIC200 · Code Composer Studio

CAD & Simulation: Fusion 360 · Abaqus (FEA) · 3D Printing (FDM)

Experience

Illini Solar Car Team

Aug 2023 – Present

Electrical Team Lead (2025–26) | Prev. Member, Electrical Team

Urbana, IL

- Electrical pit crew member of the FSGP (Formula Sun Grand Prix) 2025 #1 team.
- Characterized, tested, assembled and integrated Li-Ion battery in car's electrical system, ensuring function and high-voltage safety.
- Led solar array design and installation for fourth-generation Illini Solar Car – validated power and structural requirements.

Research

Monolithic Systems Lab, UIUC

May 2026 – Present

Undergraduate Research Assistant

Urbana, IL

- Design and implementation of a soft-robotic shower for the elderly with no lower limb mobility.
- Optimized manually-actuated soft robotic shower prototype: designed and 3D-printed custom loc-line inspired interlocking modules; fine-tuned inter-module friction through fit adjustment and surface treatment to ensure shower stability and reduce squeaking noise.

Nuclear, Plasma & Radiological Engineering (NPRE), UIUC

Jan 2025 – May 2025

Undergraduate Research Assistant

Urbana, IL

- Semester-long independent study: integration of alternative energy sources into university microgrids.
- Assessed thermal and electric requirements for simulated data centers up to 30 MW; estimated nuclear reactor energy outputs to satisfy the requirements.
- Modeled electric, absorption, and hybrid cooling strategies in Python; cut reactor thermal demand by 18% via absorption cooling.

Projects

Embedded Motor Control System — TI C2000 DSP

Aug 2026 – Present

- Developing real-time motor control firmware in C (PWM, ADC, timer ISRs); implemented optical encoder feedback for angular position/velocity at 200 Hz.
- Identified a discrete-time DC motor transfer function from step-response data
- Designing a closed-loop PI speed controller from the identified model, targeting <300 ms rise time and <1% overshoot

Image Contour Detection and Robotic Drawing Project

April 2026

- Utilized OpenCV to convert images into drawable stroke paths by skeletonizing and tracing pixel connectivity to find endpoints and junctions.
- Mapped strokes to a canvas in the world frame; executed on a UR3 over ROS1 via inverse kinematics with pen-up/down clearances.

Suitcase Handle Design Optimization

April 2026 – May 2026

- Designed an ergonomic nylon suitcase handle under structural and geometric constraints; eliminated corners in design by including fillets for manufacturing and ergonomy.
- Utilized Abaqus FEA to analyze stress concentrations across three design iterations, tuning fillet radius and wall thickness; converged on a geometry with 7+ factor of safety.